Computer Network

Rapid-Revision One-Liners

IBPS SO (IT OFFICER)

COMPLETE PREPARATION SERIES

100% SYLLABUS COVERAGE

CONCEPT-FOCUSED LEARNING

MCQ-ORIENTED APPROACH

QUICK REVISION READY

COMPILED BY

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Computer Network — One-Liners

Rapid-revision, double-layered facts for the IBPS SO (IT Officer) exam. Every line pairs a structural metric (how many / which port / what size / what complexity) with the specific descriptive items. Numbering is continuous.

Networking Basics and Data Communication

1. A data communication system has **5 components**: **message**, **sender**, **receiver**, **transmission medium**, and **protocol**.
2. There are **3 network criteria**: **performance**, **reliability**, and **security**.
3. There are **3 data-flow modes**: **simplex** (one-way), **half-duplex** (one way at a time), and **full-duplex** (both at once).
4. The classic **simplex** examples are **2**: the **keyboard** and the **monitor**; **half-duplex** example is the **walkie-talkie**; **full-duplex** example is the **telephone**.
5. The **4 goals** of networking are **resource sharing**, **reliability**, **cost saving/scalability**, and **communication medium**.
6. Network performance is measured by **2 metrics**: **throughput** (actual rate) and **delay/latency**.
7. A device qualifies for a network only if it is **autonomous** and **interconnected** — the **2** essential properties.
8. Bandwidth is measured in **hertz (Hz)** while data/bit rate is measured in **bits per second (bps)** — **2** different units.

Network Types by Scale

9. Networks by scale, smallest to largest, are **6**: **PAN**, **LAN**, **CAN**, **MAN**, **WAN**, and **GAN**.
10. A **PAN** spans about **10 metres** and is built with **Bluetooth/USB**.
11. A **MAN** spans up to about **50 km** and is exemplified by **cable TV** and city-wide networks.
12. The largest practical network is the **WAN** called the **Internet**, owned by **multiple ISPs/organizations**.
13. A **LAN** is **privately owned** and has the **highest data rate** and **lowest error rate** of the scale types.
14. A **VPN** is **1** secure logical network created by **encrypted tunneling** over a public network.
15. Two or more connected networks form **1** internetwork, and the global one is called the **Internet**.

Network Topologies

16. There are **6 basic topologies**: **bus**, **star**, **ring**, **mesh**, **tree**, and **hybrid**.
17. A full **mesh** of **n** nodes needs **n(n-1)/2** duplex links and **n-1** ports per node.
18. A **mesh** of **5** nodes needs **10** links (5×4/2) and **4** ports per node.
19. A **bus** topology needs exactly **2 terminators** at the two ends of the backbone.
20. A **star** topology with **n** nodes uses **n** cables and **n** central ports, all meeting at **1** hub/switch.
21. The **2** connection types are **point-to-point** (dedicated link) and **multipoint** (shared link).
22. The most **fault-tolerant** topology is the **mesh** (multiple paths); the least is the **bus** (single backbone).
23. A **ring** topology passes data using **1 token** and connects each node to exactly **2** neighbours.

Network Models and Service Types

24. There are **2 communication models**: **client-server** (centralized) and **peer-to-peer/P2P** (decentralized).
25. There are **2 service types**: **connection-oriented** (TCP, like a telephone) and **connectionless** (UDP/IP, like a postal letter).
26. Connection-oriented service maps to **virtual-circuit** switching and **TCP**; connectionless maps to **datagram** switching and **UDP/IP**.

Layering, OSI, and TCP/IP Models

27. The **OSI model** has **7 layers** developed by **ISO** in **1984**.
28. The **7 OSI layers** bottom-up are: **Physical, Data Link, Network, Transport, Session, Presentation, Application**.
29. The **TCP/IP model** has **4 layers**: **Application, Transport, Internet, and Network Access/Link**.
30. The **5 PDUs** by layer are: **bit** (Physical), **frame** (Data Link), **packet** (Network), **segment** (Transport-TCP), and **data/message** (upper layers).

31. The **Physical layer (L1)** transmits **bits** and defines **voltage, data rate, and topology**.

32. The **Data Link layer (L2)** uses **frames** and performs **framing, MAC addressing, error control, and flow control**.

33. The Data Link layer has **2 sublayers**: **LLC (802.2)** and **MAC**.

34. The **Network layer (L3)** uses **packets** and performs **logical (IP) addressing and routing**.

35. The **Transport layer (L4)** uses **segments** and provides **process-to-process delivery** via **port numbers**.

36. The **Session layer (L5)** provides **dialog control** and **synchronization (checkpoints)**.

37. The **Presentation layer (L6)** performs **3 jobs**: **translation, encryption/decryption, and compression**.

38. The **Application layer (L7)** provides the **user interface** and protocols like **HTTP, FTP, SMTP, DNS**.

39. The bottom **3** OSI layers are **media/support** layers; the top **3** are **user-support** layers; **Transport (L4)** is the **bridge**.

40. The **2** swapped-in-MCQs facts: **encryption/compression** is at **Presentation (L6)**, and **routing** is at **Network (L3)**.

41. There are **4 addresses** by layer: **port** (Transport), **IP/logical** (Network), **MAC/physical** (Data Link), and **name/URL** (Application).

42. The OSI bottom-up mnemonic uses **7 words**: "**Please Do Not Throw Sausage Pizza Away**".

43. TCP/IP merges OSI's **top 3 layers** (Session, Presentation, Application) into **1 Application layer**.

44. The **2** main protocols giving TCP/IP its name are **TCP** (Transport) and **IP** (Internet).

45. **OSI vs TCP/IP**: **7 vs 4** layers; **ISO vs DARPA/DoD**; **reference vs practical** model.

46. The OSI network layer supports **both** CO and CL service, but the TCP/IP **IP** layer is **connectionless only**.

47. Flow control and error control appear at **2** layers: **Data Link** (hop-to-hop) and **Transport** (end-to-end).

Physical Layer, Signals, and Channel Capacity

48. Signals are of **2 kinds**: **analog** (continuous) and **digital** (discrete).

49. Signal periodicity is of **2 types**: **periodic** (frequency f = 1/T) and **aperiodic/non-periodic**.

50. **Bit rate = baud rate × log2(L)**, where **L** is the **number of signal levels**.

51. **Bit rate is always ≥ baud rate**; they are equal only when **L = 2** (1 bit per symbol).

52. **Nyquist** bit rate (noiseless) = **2 × B × log2(L)** bits/sec.

53. **Shannon** capacity (noisy) = **B × log2(1 + S/N)** bits/sec and has **no L term**.

54. **SNR in decibels = 10 × log10(S/N)** — **1** logarithmic conversion.

55. There are **3 transmission impairments**: **attenuation** (energy loss), **distortion** (shape change), and **noise**.

56. Noise is of **4 types**: **thermal**, **induced**, **crosstalk**, and **impulse** noise.

57. **Attenuation** is measured in **decibels (dB)** and fixed using **amplifiers/repeaters**.

Data Transmission Modes

58. Transmission is of **2 forms**: **parallel** (n wires at once) and **serial** (1 wire, bit by bit).

59. Serial transmission has **3 sub-types**: **asynchronous**, **synchronous**, and **isochronous**.

60. **Asynchronous** transmission frames each byte with **1 start bit** and **1-2 stop bits**.

61. **Synchronous** transmission uses **0** start/stop bits and relies on a **shared clock**.

62. **Isochronous** transmission guarantees a **constant/fixed** data rate with **no jitter** for real-time media.

Encoding and Modulation

63. There are **4 conversion categories**: **digital-to-digital**, **digital-to-analog**, **analog-to-digital**, and **analog-to-analog**.

64. **Digital-to-digital line coding** schemes include **5**: **NRZ-L, NRZ-I, RZ, Manchester, and Differential Manchester**.

65. **NRZ-I** inverts the signal on a **1** bit and stays the same on a **0**.

66. **RZ** encoding needs **3 levels** because the signal returns to **zero** in the middle of each bit.

67. **Manchester** encoding (used in **Ethernet/802.3**) puts **1 transition** in the middle of every bit.

68. **Differential Manchester** (used in **Token Ring/802.5**) encodes a **0** by a **transition at the start** of the bit.

69. **Digital-to-analog** schemes are **4**: **ASK** (amplitude), **FSK** (frequency), **PSK** (phase), and **QAM** (amplitude+phase).

70. **QAM** packs the **most bits per symbol** of the **4** digital-to-analog schemes (e.g., 16-QAM, 64-QAM).

71. **Analog-to-digital** conversion uses **2** methods: **PCM** and **Delta Modulation**.

72. PCM has **3 steps**: **sampling**, **quantization**, and **encoding**.
73. The **Nyquist sampling theorem** requires a sampling rate of at least **2 × fmax**.
74. Delta modulation sends only **1 bit per sample** indicating an up/down change.
75. A telephone voice channel (**4 kHz**) sampled at **8000 Hz** with **8 bits** yields **64 kbps** PCM.
76. Analog-to-analog modulation has **3 types**: **AM** (amplitude), **FM** (frequency), and **PM** (phase).

Multiplexing and Spread Spectrum

77. There are **3 multiplexing techniques**: **FDM** (frequency), **TDM** (time), and **WDM** (wavelength/optical).
78. FDM separates signals using **guard bands** in the **frequency** domain (analog).
79. TDM has **2 variants**: **synchronous** (fixed slots) and **statistical/asynchronous** (on-demand slots).
80. Statistical TDM is more bandwidth-efficient than **synchronous TDM** because it allocates slots **on demand**.
81. WDM is conceptually **FDM for fiber optics**, combining beams via a **prism**.
82. Spread spectrum has **2 methods**: **FHSS** (frequency hopping) and **DSSS** (direct sequence/chip codes).

Transmission Media

83. Transmission media split into **2 classes**: **guided/wired** and **unguided/wireless**.
84. Guided media are **3**: **twisted pair**, **coaxial cable**, and **fiber optic**.
85. Twisted pair comes in **2 forms**: **UTP** (unshielded) and **STP** (shielded), using the **RJ-45** connector.
86. UTP categories: **Cat 3 = 10 Mbps**, **Cat 5 = 100 Mbps**, **Cat 5e = 1 Gbps**, and **Cat 6 = up to 10 Gbps**.
87. Coaxial cable uses the **BNC** connector and types like **RG-58, RG-59, and RG-6**.
88. Fiber optic carries **light pulses** using **total internal reflection** and is immune to **EMI**.
89. Fiber has **2 modes**: **single-mode** (laser, narrow ~**9 µm** core, long distance) and **multi-mode** (LED, ~**50-62.5 µm** core, short).
90. Fiber connectors include **4**: **SC, ST, LC, and FC**.
91. Unguided media are **3 bands**: **radio waves**, **microwaves**, and **infrared**.
92. Radio waves are **omnidirectional**, while **microwaves** are **unidirectional** and need **line-of-sight**.
93. Infrared cannot pass through walls, making it suitable only for **short-range indoor** links.
94. Of all media, **fiber optic** has the **highest bandwidth, lowest attenuation, and best EMI immunity**.
95. Of all media, **twisted pair** is the **cheapest** while **fiber** is the **most expensive**.

Switching Techniques

96. There are **3 switching techniques**: **circuit switching**, **message switching**, and **packet switching**.
97. Circuit switching has **3 phases**: **setup**, **data transfer**, and **teardown** (e.g., the **telephone/PSTN**).
98. Message switching uses **store-and-forward** with **0** dedicated paths.
99. Packet switching has **2 approaches**: **datagram** (connectionless) and **virtual circuit** (connection-oriented).
100. The **datagram** approach is used by the **Internet/IP**; **virtual circuit** is used by **X.25, Frame Relay, ATM, and MPLS**.
101. In **datagram** switching each packet carries the **full destination address**; in **virtual circuit** each carries a short **VCI**.
102. Only **2** switching techniques have a setup phase: **circuit** and **virtual-circuit** packet switching.

Data Link Layer — Framing

103. The Data Link layer performs **5 functions**: **framing, MAC addressing, error control, flow control, and access control**.
104. Framing has **3 methods**: **character count**, **byte/character stuffing**, and **bit stuffing**.
105. In **bit stuffing**, a **0** is inserted after **five consecutive 1s** in the data.
106. The HDLC flag pattern is **01111110** — **1** unique boundary marker.
107. Byte stuffing inserts an **escape (ESC)** byte before any **FLAG** or **ESC** appearing inside the data.

Error Detection and Correction

108. Errors are of **2 types**: **single-bit** (one bit flips) and **burst** (two or more bits flip).

109. **Burst errors** are the **most common** type in **serial** transmission.

110. Error-detection methods are **4**: **simple parity**, **2-D parity**, **checksum**, and **CRC**.

111. **Simple parity** detects all **odd**-numbered bit errors but **fails** for **even**-numbered errors.

112. **Two-dimensional parity** can **correct 1** single-bit error using **row and column** parity.

113. **Checksum** uses **1's complement** addition and is used at the **transport/network** layer.

114. **CRC** uses **modulo-2 (XOR)** polynomial division and is the strongest **detection** method.

115. For a generator of **degree r**, CRC appends **r zero bits** and produces an **r-bit** remainder.

116. **CRC-32** (a **4-byte** FCS) is used in **Ethernet** frames.

117. **Hamming code** corrects a **single-bit** error using parity bits at **power-of-2** positions (**1, 2, 4, 8**).

118. Hamming redundant bits **r** satisfy **2^r ≥ m + r + 1** for **m** data bits.

119. For **m = 7** data bits, Hamming needs **r = 4** parity bits (since 2^4 = 16 ≥ 12).

120. To **correct t** errors a code needs **d_min ≥ 2t + 1**; to **detect d** errors it needs **d_min ≥ d + 1**.

121. Single-error correction requires a minimum **Hamming distance of 3**.

Flow Control and ARQ

122. **ARQ** (Automatic Repeat reQuest) protocols are **3**: **Stop-and-Wait**, **Go-Back-N**, and **Selective Repeat**.

123. **Stop-and-Wait** efficiency = **1 / (1 + 2a)**, where **a = Tp/Tt**.

124. **Sliding window** efficiency = **N / (1 + 2a)**, capped at **1** (100%).

125. The optimal window size for **100% utilization** is **N = 1 + 2a**.

126. **Go-Back-N** has a sender window of **2^m – 1** and a receiver window of **1**.

127. **Selective Repeat** has sender and receiver windows both equal to **2^(m–1)**.

128. On error, **Go-Back-N** retransmits **the frame and all subsequent** frames; **Selective Repeat** retransmits **only the lost** frame.

129. **Go-Back-N** uses **cumulative** ACKs; **Selective Repeat** uses **selective/individual** ACKs.

130. Only **Selective Repeat** requires the receiver to **buffer** out-of-order frames among the **3** ARQ protocols.

131. **Piggybacking** carries **1** acknowledgement inside an outgoing **data** frame to save bandwidth.

132. **Bandwidth-Delay Product (BDP) = bandwidth × RTT** gives the number of bits "in flight".

Medium Access Control (MAC)

133. Channel-access methods fall into **3 categories**: **random access**, **controlled access**, and **channelization**.

134. **Random access** protocols are **4**: **Pure ALOHA**, **Slotted ALOHA**, **CSMA/CD**, and **CSMA/CA**.

135. **Pure ALOHA** has a maximum efficiency of **18.4%** (1/2e) at offered load **G = 0.5**.

136. **Slotted ALOHA** has a maximum efficiency of **36.8%** (1/e) at offered load **G = 1**.

137. **Slotted ALOHA doubles** the throughput of Pure ALOHA by halving the vulnerable period to **1 frame time**.

138. The vulnerable period is **2 × Tfr** for **Pure ALOHA** and **1 × Tfr** for **Slotted ALOHA**.

139. **CSMA** persistence methods are **3**: **1-persistent**, **non-persistent**, and **p-persistent**.

140. **CSMA/CD** ("listen while talking") is used in **wired Ethernet (802.3)** and uses **binary exponential backoff**.

141. **CSMA/CA** ("collision avoidance") is used in **wireless Wi-Fi (802.11)** with **RTS/CTS and IFS**.

142. **Controlled access** methods are **3**: **reservation**, **polling**, and **token passing**.

143. **Channelization** methods are **3**: **FDMA** (frequency), **TDMA** (time), and **CDMA** (codes).

IEEE 802 Standards

144. **IEEE 802.3** defines **Ethernet** using **CSMA/CD**.

145. **IEEE 802.4** defines **Token Bus** and **IEEE 802.5** defines **Token Ring**.

146. **IEEE 802.11** defines **Wireless LAN (Wi-Fi)** using **CSMA/CA**.

147. **IEEE 802.15** defines **WPAN**, including **Bluetooth (802.15.1)** and **Zigbee (802.15.4)**.

148. **IEEE 802.16** defines **WiMAX**; **IEEE 802.2** defines the **LLC** sublayer.

149. Ethernet speeds are **4**: **Standard 10 Mbps**, **Fast 100 Mbps**, **Gigabit 1 Gbps**, and **10-Gigabit 10 Gbps**.

150. Wi-Fi generations include **802.11b (11 Mbps)**, **802.11g (54 Mbps)**, **802.11n**, **802.11ac**, and **802.11ax (Wi-Fi 6)**.

Ethernet Frame and MAC Addressing

151. The Ethernet frame has a **7-byte preamble** and a **1-byte SFD (10101011)**.

152. The Ethernet minimum frame is **64 bytes** and the maximum is **1518 bytes**.

153. The Ethernet payload/MTU is **46-1500 bytes**, with **46** the minimum data size (padded).

154. A **MAC address** is **48 bits = 6 bytes = 12 hex digits**, e.g., `00:1A:2B:3C:4D:5E`.

155. The first **24 bits** of a MAC address are the **OUI** (manufacturer identifier).

156. The broadcast MAC address is **FF:FF:FF:FF:FF:FF** — all **48** bits set to **1**.

157. A Bluetooth **piconet** has **1 master** and up to **7 active slaves** (plus 255 parked).

158. Bluetooth uses **FHSS** in the **2.4 GHz** band — **1** unlicensed ISM band.

Network Layer Functions

159. The Network layer performs **4 functions**: **logical addressing, routing, packetizing, and fragmentation**.

160. **Routing** builds the path (control plane) while **forwarding** moves a packet to the right port (data plane) — **2** distinct jobs.

161. The IP service model has **2** properties: **connectionless** and **best-effort** (unreliable).

IPv4 Addressing

162. An **IPv4 address** is **32 bits**, written as **4 octets** in **dotted-decimal**, each **0-255**.

163. The total IPv4 space is **2^32 ≈ 4.29 billion** addresses.

164. IPv4 has **5 classes**: **A, B, C, D, and E**.

165. **Class A** uses first octet **0-127**, default mask **/8**, giving **126** networks and **2^24–2** hosts.

166. **Class B** uses first octet **128-191**, default mask **/16**, giving **2^14** networks and **2^16–2** hosts.

167. **Class C** uses first octet **192-223**, default mask **/24**, giving **2^21** networks and only **254** hosts.

168. **Class D** (**224-239**) is for **multicast**; **Class E** (**240-255**) is **reserved/experimental**.

169. The leading bits are **0** (A), **10** (B), **110** (C), **1110** (D), and **1111** (E).

170. Usable hosts per network = **2^h – 2**, subtracting the **network** and **broadcast** addresses.

171. The **loopback** address is **127.0.0.1**, part of the **127.0.0.0/8** block.

172. **APIPA** uses the **169.254.0.0/16** range when a **DHCP** server is unavailable.

173. The **3 private ranges** are **10.0.0.0/8**, **172.16.0.0/12**, and **192.168.0.0/16**.

174. The limited broadcast address is **255.255.255.255** and the default route is **0.0.0.0**.

Subnetting, CIDR, and Supernetting

175. **Subnetting** creates **2^n** subnets by borrowing **n** host bits.

176. **Supernetting** is the **opposite** of subnetting, merging networks into **1** larger block.

177. **CIDR** drops class boundaries and uses a **/n prefix**, e.g., `192.168.1.0/26`.

178. Block size (increment) = **256 – mask octet value**; for **/26** it is **256 – 192 = 64**.

179. A **/26** network has **6** host bits, giving **62** usable hosts (2^6 – 2).

180. A **/30** subnet gives exactly **2** usable hosts, ideal for **point-to-point** links.

181. A **/31** is special (RFC 3021) giving **2** addresses with **0** wasted for point-to-point links.

182. Subnetting has **2 styles**: **FLSM** (fixed-length) and **VLSM** (variable-length, more efficient).

183. The network address has host bits all **0**; the broadcast address has host bits all **1**.

IPv4 Datagram and Supporting Protocols

184. The **IPv4 header** is **20 bytes minimum** and **60 bytes maximum** (with options).

185. The IPv4 **Total Length** field is **16 bits**, allowing a maximum datagram of **65,535 bytes**.

186. The IPv4 **TTL** field is **8 bits** and is **decremented by 1** at each router (discarded at **0**).

187. Common IPv4 **Protocol** numbers: **ICMP = 1**, **IGMP = 2**, **TCP = 6**, **UDP = 17**, and **OSPF = 89**.

188. IPv4 **fragmentation** uses **3 fields**: **Identification**, **Flags (DF/MF)**, and **Fragment Offset**.

189. IPv4 reassembly happens only at the **destination**, and the fragment offset is measured in units of **8 bytes**.

190. **ARP** resolves **IP → MAC** while **RARP** resolves **MAC → IP** — **2** opposite mappings.

191. **ICMP** powers **2** key tools: **ping** (echo request/reply) and **traceroute** (time exceeded).

192. **IGMP** manages **multicast** group membership at the **Network layer**.

193. Common ICMP messages include **Destination Unreachable**, **Time Exceeded**, and **Echo Request/Reply**.

194. **NAT** maps **private to public** IPs; **PAT (overloading)** shares **1** public IP among many hosts using **port numbers**.

195. **DHCP** uses the **DORA** process: **Discover, Offer, Request, Acknowledge** — **4** steps.

196. **DHCP** runs on **UDP** ports **67 (server)** and **68 (client)**.

IPv6

197. An **IPv6 address** is **128 bits**, written as **8 groups** of **16-bit hex** separated by colons.

198. The total IPv6 space is **2^128** addresses — vastly larger than IPv4's **2^32**.

199. The **IPv6 header** is a **fixed 40 bytes**, unlike IPv4's variable **20-60 bytes**.

200. IPv6 has **3 address types**: **unicast**, **multicast**, and **anycast** — with **no broadcast**.

201. IPv6 removes **2** things present in IPv4: the **header checksum** and **router-based fragmentation**.

202. In IPv6, only the **source** node may fragment, not intermediate routers.

203. IPv6 zero-compression replaces consecutive zero groups with **::**, usable only **once** per address.

204. IPv6 auto-configuration uses **SLAAC** or **DHCPv6** — **2** methods.

205. IPv6 transition mechanisms are **3**: **dual stack**, **tunneling**, and **translation (NAT64)**.

Routing

206. Routing protocols use **3 algorithm families**: **Distance Vector**, **Link State**, and **Path Vector**.

207. **Distance Vector** uses the **Bellman-Ford** algorithm; **Link State** uses **Dijkstra**.

208. **RIP** is a **Distance Vector** protocol using **hop count**, with a maximum of **15 hops** (**16 = infinity**).

209. **RIP** sends full-table updates every **30 seconds** over **UDP port 520**.

210. **OSPF** is a **Link State** protocol using **cost** metric and runs directly over IP as **protocol 89**.

211. **BGP** is the **Path Vector** inter-domain protocol of the Internet, running on **TCP port 179**.

212. The **count-to-infinity** problem affects **Distance Vector** routing only.

213. Count-to-infinity is mitigated by **3 techniques**: **split horizon**, **poison reverse**, and **hold-down timers**.

214. **Link State** routers flood **LSAs** to **all** routers and build a complete **topology map**.

215. **Distance Vector** converges **slowly** while **Link State** converges **fast** — **2** opposite behaviours.

216. **Flooding** forwards a packet on **every** outgoing line except the one it arrived on.

217. Routing is of **2 modes**: **static** (manual) and **dynamic** (auto via protocols).

Transport Layer — Ports, UDP, and TCP

218. Port numbers are **16 bits**, ranging **0-65535** — about **65,536** values.

219. Ports have **3 ranges**: **well-known (0-1023)**, **registered (1024-49151)**, and **dynamic/ephemeral (49152-65535)**.

220. A **socket** is the pair **IP address + port number** — **2** parts identifying an endpoint.

221. The Transport layer offers **2 protocols**: **TCP** (reliable, connection-oriented) and **UDP** (unreliable, connectionless).

222. The **UDP header** is a fixed **8 bytes** with **4 fields**: **source port, destination port, length, checksum**.

223. **UDP** is used by **DNS, DHCP, TFTP, SNMP, VoIP, and streaming** for low latency.

224. The **TCP header** is **20-60 bytes**, with **6** primary control flags.

225. The **6 TCP flags** are **URG, ACK, PSH, RST, SYN, and FIN**.

226. TCP **Sequence** and **Acknowledgement** numbers are each **32 bits**.

227. The TCP **Window Size** field is **16 bits** and is used for **flow control**.

228. TCP connection setup is a **3-way handshake**: **SYN → SYN+ACK → ACK**.

229. TCP connection termination is a **4-way** exchange: **FIN → ACK → FIN → ACK**.

230. The **SYN** flag synchronizes sequence numbers; the **RST** flag abruptly resets a connection.

TCP Flow and Congestion Control

231. TCP's effective window is **min(rwnd, cwnd)** — the smaller of **2** windows.

232. TCP congestion control has **4 phases**: **Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery**.

233. In **Slow Start**, cwnd grows **exponentially**, doubling every **RTT** until **ssthresh**.

234. In **Congestion Avoidance**, cwnd grows **linearly** (**+1 MSS per RTT**).

235. **Fast Retransmit** triggers on **3 duplicate ACKs** without waiting for a timeout.

236. TCP's overall policy is **AIMD** — **Additive Increase, Multiplicative Decrease**.

237. Traffic shaping uses **2 algorithms**: **Leaky Bucket** (constant rate) and **Token Bucket** (allows bursts).

238. The **Leaky Bucket** outputs at a **fixed** rate, while the **Token Bucket** permits bursts using saved **tokens**.

239. **QoS** is defined by **4 parameters**: **bandwidth, delay/latency, jitter, and packet loss**.

240. **TCP vs UDP**: TCP is **reliable/ordered** with a **20-60 B** header; UDP is **unreliable/fast** with an **8 B** header.

Application Layer — DNS

241. **DNS** translates **domain names to IP addresses** using **port 53** (UDP for queries, TCP for zone transfers).

242. DNS hierarchy has **4 levels**: **root (.)** → **TLD** → **second-level** → **host**.

243. DNS resolution has **2 styles**: **recursive** (resolver does all work) and **iterative** (referrals).

244. Key DNS records: **A** (IPv4), **AAAA** (IPv6), **MX** (mail), **CNAME** (alias), **NS** (name server), **PTR** (reverse).

245. A **PTR** record does **reverse** lookup (IP → name); an **A** record does forward lookup (name → IPv4).

Application Layer — Protocols and Ports

246. **FTP** uses **2 ports**: **20 (data)** and **21 (control)**, both over **TCP**.

247. **SSH** uses **port 22** and is the secure replacement for **Telnet** (port **23**, insecure).

248. **SMTP** uses **port 25** to **send/push** email.

249. Email retrieval uses **2 protocols**: **POP3 (port 110)** and **IMAP (port 143)**.

250. **POP3** typically **downloads and deletes** mail; **IMAP** keeps mail on the **server** for multi-device sync.

251. **HTTP** uses **port 80**; **HTTPS** uses **port 443** (HTTP over **TLS/SSL**).

252. **DNS** uses **port 53**; **DHCP** uses **ports 67/68**; **TFTP** uses **port 69**.

253. **SNMP** uses **ports 161/162** (port **162** for **traps**) for network management.

254. **NTP** uses **port 123** for time synchronization; **BGP** uses **TCP port 179**.

255. **HTTP** is **stateless**, and state is maintained using **cookies**.

256. HTTP defines methods such as **GET, POST, PUT, DELETE, and HEAD** — at least **5** common verbs.

257. **TFTP** (port 69) runs over **UDP**, while regular **FTP** runs over **TCP**.

258. **SMTP is push**, while **POP3/IMAP are pull** — **2** opposite email transfer directions.

World Wide Web and Internet

259. A **URL** has the form `protocol://host:port/path`, combining **4** parts.

260. The **WWW** uses **HTTP/HTTPS** over the Internet and links **hypertext** documents.

261. A **proxy server** performs **3** roles: **caching**, **filtering**, and improving **privacy/performance**.

262. **Cookies** are small browser-stored data used to add **state** to the **stateless** HTTP protocol.

Networking Devices and OSI Mapping

263. A **repeater** operates at **Layer 1 (Physical)** and **regenerates** weak signals to extend distance.

264. A **hub** operates at **Layer 1**, broadcasts to **all** ports, and forms **1** collision domain.

265. A **bridge** operates at **Layer 2 (Data Link)** and filters traffic by **MAC** address.

266. A **switch** operates at **Layer 2** (L3 switch at Layer 3) and gives each port its **own** collision domain.

267. A **router** operates at **Layer 3 (Network)**, forwards by **IP**, and separates **broadcast** domains.

268. A **gateway** operates at **Layer 7 / all layers** and acts as a **protocol converter**.

269. A **modem** operates at **Layer 1** and performs **modulation/demodulation** (digital ↔ analog).

270. A **brouter** is a hybrid working at **Layers 2 and 3** (bridge + router).

271. A **NIC** works at **Layers 1-2** and carries the burned-in **48-bit MAC** address.

272. The number of collision domains equals the number of **switch ports**, but a **hub** has just **1**.

273. A **switch** separates **collision** domains but **not** broadcast domains; a **router** separates **both**.

274. **Layer-1** devices are **3**: **hub, repeater, and modem**.

275. **Layer-2** devices are **3**: **switch, bridge, and NIC**.

276. A **VLAN** (802.1Q) logically segments **1** physical switch into multiple **broadcast** domains.

277. **Hub vs switch**: a hub is **half-duplex** with shared bandwidth; a switch is **full-duplex** with dedicated bandwidth.

Network Security

278. The **CIA triad** has **3 goals**: **Confidentiality**, **Integrity**, and **Availability**.

279. Cryptography has **2 main types**: **symmetric** (one key) and **asymmetric** (public/private key pair).

280. **Symmetric** algorithms include **3**: **DES (56-bit), 3DES, and AES (128/192/256-bit)**.

281. **Asymmetric** algorithms include **RSA, Diffie-Hellman, and ECC** — at least **3**.

282. **Symmetric** encryption is **fast** but has a **key-distribution** problem that **asymmetric** solves.

283. **Diffie-Hellman** is a key **exchange** algorithm, not an encryption algorithm.

284. Common hash functions: **MD5 (128-bit)**, **SHA-1 (160-bit)**, and **SHA-256** — **3** one-way digests.

285. A **digital signature** provides **3 services**: **authentication, integrity, and non-repudiation**.

286. To **sign**, use the sender's **private** key; to **verify**, use the sender's **public** key — **2** opposite keys.

287. For **confidentiality**, encrypt with the **receiver's public** key and decrypt with the **receiver's private** key.

288. **Firewalls** have **3 types**: **packet-filter**, **stateful inspection**, and **application/proxy gateway**.

289. A **VPN** creates **1** encrypted tunnel over a public network using **IPSec** or **SSL/TLS**.

290. A **DoS/DDoS** attack targets **availability**, the **A** in the CIA triad.

291. Common attacks include **DoS, MITM, phishing, spoofing, sniffing, and replay** — at least **6** categories.

292. **Hashing** is **one-way/irreversible**, while **encryption** is **two-way/reversible** — **2** opposite properties.

293. **TLS/SSL** typically uses **asymmetric** crypto to exchange a **symmetric** session key (a hybrid of **2** schemes).

Additional High-Yield Facts — Delays, Signals, and Coding

294. End-to-end delay has **4 components**: **transmission, propagation, queuing, and processing** delay.

295. **Transmission delay = packet size / bandwidth**, while **propagation delay = distance / propagation speed**.

296. Power gain/loss in decibels = **10 × log10(P2/P1)** — **1** standard formula.

297. **Manchester** encoding needs a baud rate **twice** the bit rate, so it uses **2×** the bandwidth.

298. Scrambling techniques are **2** regional: **B8ZS** (North America/T1) and **HDB3** (Europe/E1).

299. Block coding schemes include **4B/5B** (FDDI/100BASE-FX) and **8B/10B** (Gigabit Ethernet).

300. **Bipolar AMI** uses **3 levels** where successive **1s** alternate in polarity around a **0** level.

301. A **T1** line carries **24** voice channels at **1.544 Mbps**; an **E1** line carries **32** channels at **2.048 Mbps**.

302. OSI's **7 layers** have **6** interfaces between adjacent layers.

303. During encapsulation every layer adds a **header**, but only the **Data Link** layer also adds a **trailer**.

Additional Facts — Switching and WAN

304. An **ATM cell** is a fixed **53 bytes**: a **5-byte header** plus a **48-byte payload**.

305. Connection-oriented WAN technologies include **3**: **X.25**, **Frame Relay**, and **ATM**.

306. **Frame Relay** identifies virtual circuits using a **DLCI**; **ATM** uses a **VPI/VCI** pair.

307. **MPLS** forwards using a **20-bit label**, operating between **Layer 2 and Layer 3** ("Layer 2.5").

Additional Facts — Data Link Protocols

308. **HDLC** is a **bit-oriented** protocol using the flag **01111110** and **bit stuffing**.

309. **PPP** is a **byte-oriented** protocol that uses **byte stuffing** and has **2** sub-protocols: **LCP** and **NCP**.

310. HDLC defines **3 frame types**: **I-frames** (information), **S-frames** (supervisory), and **U-frames** (unnumbered).

311. With **m** sequence-number bits, the total sequence space is **2^m** distinct numbers.

312. CRC standards include **3** common widths: **CRC-16**, **CRC-CCITT (16-bit)**, and **CRC-32 (32-bit)**.

Additional Facts — Wireless MAC

313. **IEEE 802.11** defines **3 inter-frame spaces** by priority: **SIFS < PIFS < DIFS**.

314. **RTS/CTS** handshaking in **802.11** solves the **hidden terminal** problem using **2** control frames.

315. The **2** key wireless problems are the **hidden terminal** and the **exposed terminal** problems.

316. Wi-Fi security has evolved through **4** schemes: **WEP, WPA, WPA2, and WPA3**.

Additional Facts — IP and IPv6

317. The IPv4 **Options** field can be up to **40 bytes**, taking the header from **20** to **60** bytes.

318. The default subnet masks are **3**: **/8** (Class A), **/16** (Class B), and **/24** (Class C).

319. A **/8** mask is **255.0.0.0**, a **/16** is **255.255.0.0**, and a **/24** is **255.255.255.0**.

320. The IPv6 **loopback** address is **::1** and the **unspecified** address is **::**.

321. The IPv6 **link-local** prefix is **fe80::/10**, and the **multicast** prefix is **ff00::/8**.

322. A **/64** prefix is the standard IPv6 subnet size, leaving **64 bits** for the interface ID.

323. **ICMPv6** combines the roles of **ICMP, ARP, and IGMP** from IPv4 — merging **3** functions.

324. **MTU** of standard Ethernet is **1500 bytes**, while **jumbo frames** reach up to **9000 bytes**.

325. An **802.1Q VLAN tag** adds **4 bytes**, raising the maximum Ethernet frame to **1522 bytes**.

Additional Facts — Transport Layer

326. The TCP **TIME-WAIT** state lasts **2 × MSL** (maximum segment lifetime) before closing.

327. The TCP **Options** field can be up to **40 bytes**, extending the header from **20** to **60** bytes.

328. **Nagle's algorithm** reduces the number of small packets by buffering until an ACK or full segment.

329. **Silly Window Syndrome** is avoided by **2** rules: **Clark's solution** (receiver) and **Nagle's** (sender).

330. The **MSS** (Maximum Segment Size) for standard Ethernet TCP is typically **1460 bytes** (1500 – 20 – 20).

331. TCP has **3** congestion-detection signals: a **timeout**, **3 duplicate ACKs**, and (with ECN) an **explicit notification**.

332. The TCP **window-scaling** option allows windows larger than the base **65,535-byte** limit.

Additional Facts — Application Layer

333. **HTTP status codes** have **5 classes**: **1xx** info, **2xx** success, **3xx** redirect, **4xx** client error, **5xx** server error.

334. Common HTTP codes: **200 OK**, **301/302 redirect**, **400 Bad Request**, **401 Unauthorized**, **403 Forbidden**, **404 Not Found**, **500 Internal Server Error**.

335. **HTTP versions** include **5**: **0.9, 1.0, 1.1, 2.0, and 3.0**.

336. **HTTP/3** runs over **QUIC**, which uses **UDP** instead of TCP.

337. **HTTP/1.1** introduced **persistent connections** (keep-alive) and **pipelining**.

338. **MIME** extends **SMTP** to carry **non-text** content like images and attachments.

339. **DNS** has **13 logical root server** clusters, labelled **A through M**.

340. **TLDs** are of **2 main kinds**: **gTLD** (.com, .org) and **ccTLD** (.in, .uk).

341. A **URI** is the superset; a **URL** (location) and a **URN** (name) are its **2** subtypes.

342. **FTP** runs in **2 modes**: **active** (server opens data connection) and **passive** (client opens it).

Additional Facts — Network Security

343. **DES** uses a **64-bit block** and an effective **56-bit key**, run over **16 rounds**.

344. **AES** uses a **128-bit block** and key sizes of **128, 192, or 256 bits**.

345. **RSA** security relies on the difficulty of **factoring large integers** (product of two primes).

346. **IPSec** has **2 protocols**: **AH** (Authentication Header) and **ESP** (Encapsulating Security Payload).

347. **IPSec** operates in **2 modes**: **transport mode** and **tunnel mode**.

348. **Kerberos** is a **ticket-based** authentication protocol using a trusted **KDC** (Key Distribution Center).

349. **SSL/TLS** secures HTTPS on **port 443** and sits between the **Transport** and **Application** layers.

350. A **CA (Certificate Authority)** issues **digital certificates** binding a **public key** to an identity in a **PKI**.

351. **MAC (Message Authentication Code)** provides **integrity + authentication** using a **shared secret key**.

352. **DDoS** attacks use many **compromised hosts (a botnet)** to overwhelm **1** target's availability.

Rapid Recap — Most-Tested Numbers

353. OSI = **7 layers**, TCP/IP = **4 layers**, and the difference between them is **3 layers**.

354. Address sizes: **IPv4 = 32 bits**, **IPv6 = 128 bits**, **MAC = 48 bits**, and **Port = 16 bits**.

355. Header sizes: **UDP = 8 B**, **TCP = 20-60 B**, **IPv4 = 20-60 B**, and **IPv6 = 40 B fixed**.

356. ALOHA efficiencies: **Pure = 18.4%** and **Slotted = 36.8%** — a **2×** improvement.

357. RIP limits: **max 15 hops**, **infinity = 16**, **30-second** updates, on **UDP 520**.

358. Routing protocol ports: **RIP = UDP 520**, **BGP = TCP 179**, and **OSPF = IP protocol 89**.

359. Email ports: **SMTP = 25**, **POP3 = 110**, and **IMAP = 143**.

360. Web ports: **HTTP = 80** and **HTTPS = 443**, differing by the **TLS/SSL** encryption layer.

361. Secure vs insecure remote login: **SSH = 22** (secure) vs **Telnet = 23** (plaintext).

362. ARQ windows: **Go-Back-N = 2^m − 1** and **Selective Repeat = 2^(m−1)**.

363. Channel capacity: **Nyquist = 2B·log2(L)** (noiseless) and **Shannon = B·log2(1+S/N)** (noisy).

364. Device layers: **Hub/Repeater = L1**, **Switch/Bridge/NIC = L2**, **Router = L3**, **Gateway = L7**.

365. Ethernet frame: **min 64 B**, **max 1518 B**, **MTU 1500 B**, with a **4-byte CRC-32** FCS.

366. Subnetting: **subnets = 2^n**, **hosts = 2^h − 2**, **block size = 256 − mask octet**.

More Atomic Facts — Physical and Media

367. **16-QAM** carries **4 bits/symbol**, **64-QAM** carries **6 bits/symbol**, and **256-QAM** carries **8 bits/symbol**.

368. Coaxial impedance is **2** values: **50 ohm** (thin Ethernet) and **75 ohm** (cable TV).

369. Fiber operates at **3 wavelength windows**: **850 nm**, **1310 nm**, and **1550 nm**.

370. **Single-mode** fiber typically uses **1310/1550 nm lasers**; **multi-mode** uses **850 nm LEDs**.

More Atomic Facts — Ethernet and LAN

371. Classic Ethernet uses a **slot time of 512 bit-times**, equal to the **64-byte** minimum frame.

372. **100BASE-TX** Fast Ethernet uses **2 pairs** of Cat5, while **1000BASE-T** Gigabit uses all **4 pairs**.

373. Ethernet naming decodes as **speed-modulation-medium**, e.g., **10BASE-T = 10 Mbps baseband twisted pair**.

374. A collision domain on a hub forces **half-duplex**, so only **1** device transmits successfully at a time.

More Atomic Facts — IP Addressing Details

375. A **/24** network provides **256** total and **254** usable host addresses.

376. The usable **Class A** network range is **1-126**, since **0** is reserved and **127** is loopback.

377. Network bits/host bits per class: **A = 8/24**, **B = 16/16**, and **C = 24/8**.

378. **CIDR aggregation** combines contiguous blocks; e.g., four **/24s** can become one **/22**.

379. The **all-zeros** host part is the **network ID** and the **all-ones** host part is the **directed broadcast**.

More Atomic Facts — Routing Protocols

380. **RIPv1** is **classful** while **RIPv2** is **classless** and supports **VLSM/CIDR**.

381. **OSPF** uses **area 0** (0.0.0.0) as the mandatory **backbone** area in a hierarchical design.

382. **EIGRP** is Cisco's **advanced/hybrid distance-vector** protocol using the **DUAL** algorithm.

383. Routing protocols split into **2 scopes**: **IGP** (RIP, OSPF, EIGRP) inside an AS and **EGP** (BGP) between ASes.

384. Common administrative distances: **OSPF = 110** and **RIP = 120**, so OSPF is preferred over RIP.

385. OSPF exchanges **LSAs** and elects a **DR** and **BDR** on multi-access networks — **2** designated roles.

More Atomic Facts — Application Ports and Protocols

386. **LDAP** uses **port 389** for directory services and **636** for **LDAPS** (secure).

387. **RDP** (Remote Desktop) uses **port 3389**, and **SMB** (file sharing) uses **port 445**.

388. Database default ports: **MySQL = 3306** and **PostgreSQL = 5432**.

389. **Telnet** uses an **NVT** (Network Virtual Terminal) abstraction over **port 23**.

390. **Syslog** uses **UDP port 514**, and **Kerberos** uses **port 88**.

391. **DHCP** assigns addresses for a finite **lease** time, after which the client must **renew**.

More Atomic Facts — ICMP, TTL, and Diagnostics

392. **ICMP echo request** is **type 8** and **echo reply** is **type 0** — the **2** messages used by **ping**.

393. Default initial **TTL** values: **Windows = 128**, **Linux/Unix = 64**, and many routers = **255**.

394. **traceroute** works by sending packets with **increasing TTL** values to map each hop.

395. An **ICMP Time Exceeded** message is generated when **TTL** reaches **0** at a router.

More Atomic Facts — Cryptography and Security

396. **RSA** was published in **1977** by **Rivest, Shamir, and Adleman** — **3** inventors.

397. **Diffie-Hellman** key exchange was published in **1976**, predating RSA by **1** year.

398. **SHA-2** family digests include **SHA-256** (256-bit) and **SHA-512** (512-bit) outputs.

399. **MD5** produces a **128-bit** digest but is **broken** (collisions found) for security use.

400. A **PKI** binds identities to public keys using **digital certificates** issued by a **CA**.

401. **WPA2** uses **AES-CCMP** encryption, while the older **WEP** used the weak **RC4** cipher.

402. **Symmetric** crypto needs **$n(n-1)/2$** keys for **n** users, whereas **asymmetric** needs only **$2n$** keys.

Final Numeric Snapshot

403. The **3** highest-yield difference tables are **OSI vs TCP/IP**, **TCP vs UDP**, and **Distance Vector vs Link State**.

404. The **4** emphasis numericals are **subnetting**, **sliding-window efficiency**, **CRC**, and **Hamming code**.

405. **PCM** for telephone voice yields **64 kbps** from **8000 samples/sec × 8 bits**.

406. A full-mesh of **$n = 10$** nodes needs **45** links (10×9/2) and **9** ports per node.

407. The **5** OSI PDUs in order are **Data, Segment, Packet, Frame, Bit** from Application down to Physical.

408. A **/30** subnet (mask **255.255.255.252**) wastes **2** of its **4** addresses on network and broadcast.